# Cytotoxic immune infiltration effects on breast cancer survival and immunotherapy response

## Cross-cohort analysis from METABRIC, TCGA-BRCA & I-SPY2 trial

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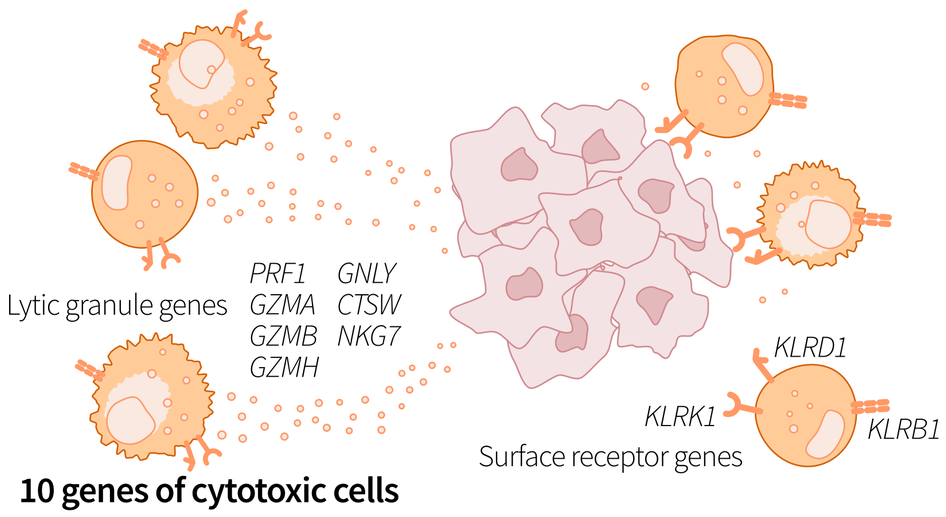
**3** cohorts — METABRIC[1], TCGA-BRCA[2,3], I-SPY2 trial[4]

**14** immune cell populations[5]

**57** harmonised shared genes[5]




- **Danaher et al. developed a panel of immune population marker genes[4] to score immune infiltration from bulk gene expression.**
- **We screened immune cell populations for associations with survival in two cohorts and with immunotherapy response in a third.**

## Identifying representative immune population across two cohorts

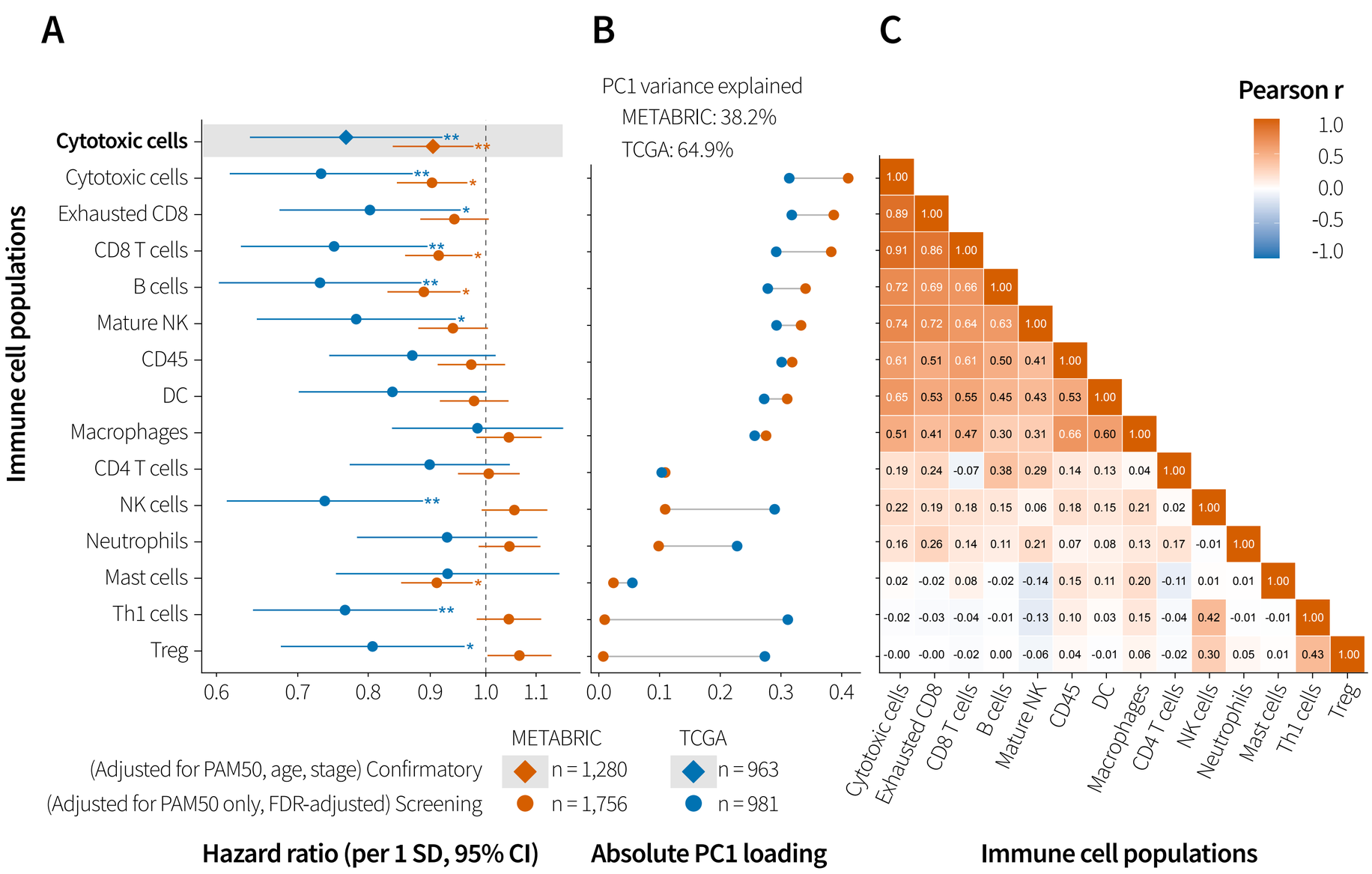



**Figure 1 | Identification of cytotoxic cells as a cross-cohort immune axis in breast cancer.**
**(a)** Hazard ratios for 14 immune populations from Cox models in METABRIC and TCGA-BRCA; top row shows fully adjusted cytotoxic result.
**(b)** Absolute PC1 loadings per immune population across both cohorts.
**(c)** Pairwise Pearson correlations between population scores in METABRIC, ordered by PC1 loading.

## Methods & data sources

**METABRIC[1]** `cBioPortalData`[6-8]
2,509 clinical
↓ − 529 no expression
1,980
↓ − 244 no OS / PAM50
1,756 screening
↓ − 476 no age / stage
1,280 confirmatory

**TCGA-BRCA[2,3]** `cBioPortalData`[6-8]
1,084 clinical
↓ − 2 no expression
1,082
↓ − 101 no OS / PAM50
981 screening
↓ − 18 no age / stage
963 confirmatory

**I-SPY2[4]** - GSE194040 `GEOquery`[9,10]
988 profiled
↓ − 740 other trial arms
248 response
↓ 69 pembrolizumab ↓ 179 control

**Cox assumptions**
Proportional hazards
- Schoenfeld residual test

Linearity
- Martingale residual test

See bottom QR code for supp info

**Immune population scoring**
Danaher et al.: 57 genes, 14 populations[5]

## Association with survival and response

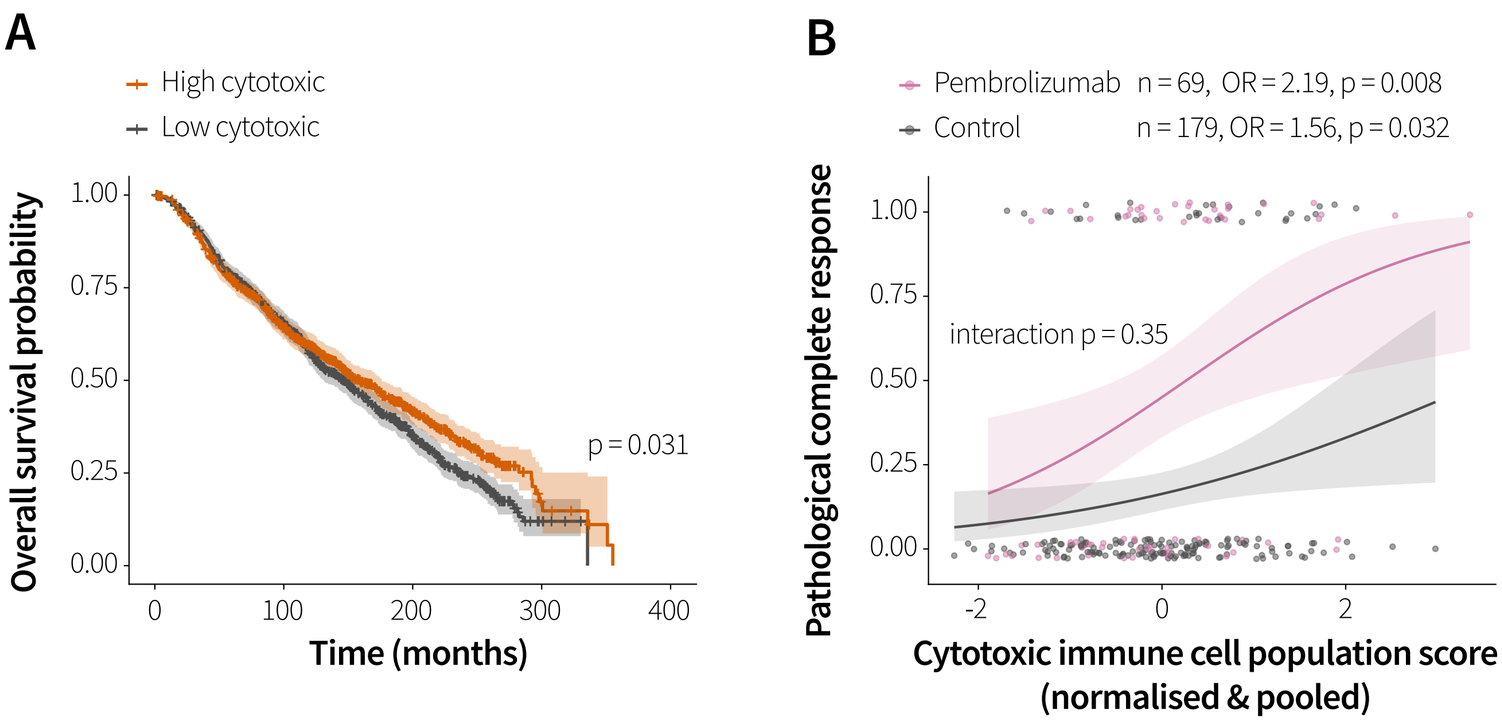
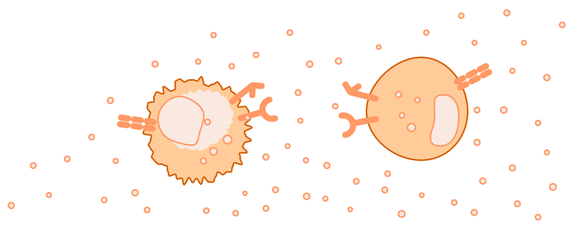



**Figure 2 | Cytotoxic infiltration associates with survival and immunotherapy response.**
**(a)** Unadjusted Kaplan–Meier overall survival in METABRIC by median split of cytotoxic score.
**(b)** Logistic regression of pathological complete response against pooled cytotoxic score in the I-SPY2 pembrolizumab & paclitaxel arm and paclitaxel-only control arm.
All shaded bands are 95% CI.

## Cytotoxic infiltration links prognosis to immunotherapy benefit

- **Across METABRIC and TCGA-BRCA, higher cytotoxic scores were independently associated with better overall survival after adjusting for PAM50, age at diagnosis, and tumour stage.**
- **In I-SPY2, higher pre-treatment cytotoxic infiltration associates with pathological complete response in both arms, with a larger point estimate under pembrolizumab than paclitaxel alone.**



**References**
**Supplementary information**
**Repository**
joelyu.github.io/HDS-W4/HDS_04_YuChungYan_2604.html